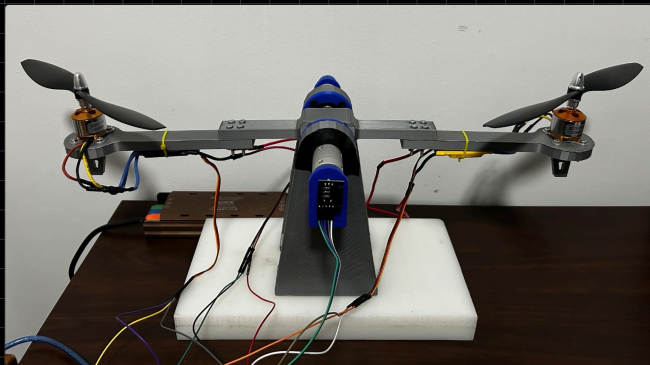
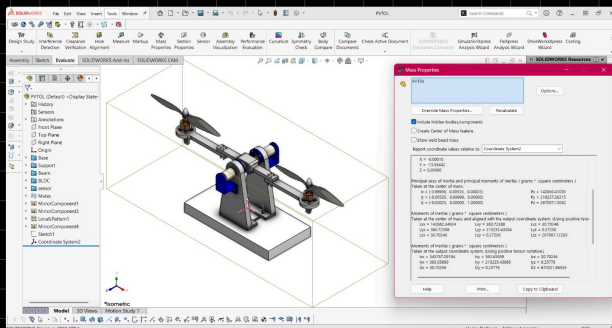


P-VTOL Project Documentation



Mechanical Design

The model was designed with the aid of SolidWorks, as shown below.



Components :

2 BLDC Motors (2212-6T-220KV)

2 Propellers (1045)

2 30A ESCs

2 Bearings (6201)

2 Flexible Couplings (6.35 x 10 L30)

1 Potentiometer (Single-Turn)

1 Rotary Encoder (KY-040)

1 IMU (MPU6050)

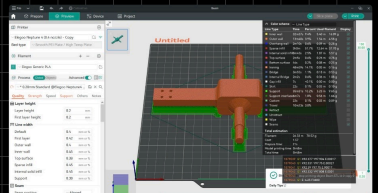
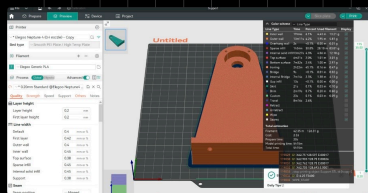
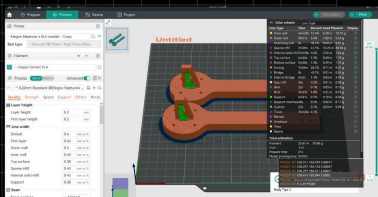
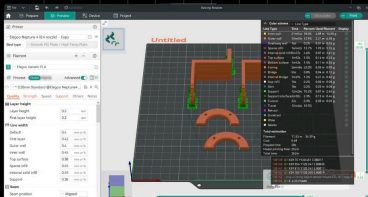
Replacable
with IMU

Bolts & Nuts.

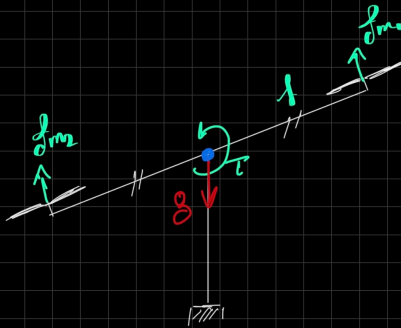
1 (30 x 15 x 3) cm Teflon Sheet

Fabrication

Test-Rig Parts were 3D-Printed using
Elegoo Neptune 4 with the following
Print Settings.



Mathematical Modeling



Real System is fixed .. Restoring Gravity Force not Present !

Lagrange Method $q = \theta$



$$T = \frac{1}{2} J \dot{\theta}^2$$

$$U = 0$$

$$L = T - U \Rightarrow L = \frac{1}{2} J \dot{\theta}^2$$

External Torques $\rightarrow u \leftarrow$ Controller
 $\rightarrow T_d \leftarrow$ Damping

$$\therefore Q = u - b\dot{\theta}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = Q$$

$$I = \frac{1}{2} J \dot{\theta}^2 \quad \frac{\partial L}{\partial \dot{\theta}} = J \dot{\theta}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = J \ddot{\theta} \quad \frac{\partial L}{\partial \theta} = 0$$

$$J \ddot{\theta} + b \dot{\theta} = u \quad u = 1(\dot{\phi}_{m1} - \dot{\phi}_{m2})$$

Newton's Method:

$$\mathcal{J} \ddot{\theta} = J \ddot{\theta}$$



$$\mathcal{L}_3 = 0 \quad \begin{array}{l} \nearrow \text{cm @ Pivot} \\ \rightarrow \text{equal distance} \end{array}$$

$$\mathcal{L}_d = b \dot{\theta}$$

$$\mathcal{L}_m = 1(\dot{\phi}_{m1} - \dot{\phi}_{m2})$$

$$J \ddot{\theta} = 1(\dot{\phi}_{m1} - \dot{\phi}_{m2}) - b \dot{\theta}$$

$$\ddot{\theta} = \frac{1}{J} (\dot{\phi}_{m1} - \dot{\phi}_{m2}) - \frac{b}{J} \dot{\theta}$$

State Space Representation

$$x_1 = \theta$$

$$\dot{x}_1 = \dot{\theta} = x_2$$

$$x_2 = \dot{\theta}$$

$$\begin{aligned}\dot{x}_2 = \ddot{\theta} &= \frac{1}{J} (\tau_m - \tau_m) - \frac{b}{J} x_2 \\ &= \frac{1}{J} u - \frac{b}{J} x_2\end{aligned}$$

$$\begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{Bmatrix} = \begin{Bmatrix} 0 & 1 \\ 0 & -\frac{b}{J} \end{Bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} + \begin{Bmatrix} 0 \\ \frac{1}{J} \end{Bmatrix} u$$

$$y = \begin{Bmatrix} 1 & 0 \\ 0 & 1 \end{Bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

System's Parameters:

$$l = 22 \text{ cm}$$

$$= 0.22 \text{ m}$$

$$b = ?$$

$$= 0.02$$

$$\text{N.m.s/rod}$$

$$J = ?$$

$$= 0.067 \text{ kg/m}^2$$

(Obtained from
Solidworks)

$$A = \begin{Bmatrix} 0 & 1 \\ 0 & -0.3 \end{Bmatrix}$$

$$C = \begin{Bmatrix} 1 & 0 \\ 0 & 1 \end{Bmatrix}$$

$$B = \begin{Bmatrix} 0 \\ 15 \end{Bmatrix}$$

$$D = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Control System Design

$$\text{Ch. eqn.} \Rightarrow \det(sI - A) = 0$$

$$\left| \begin{Bmatrix} s & -1 \\ 0 & s+0.3 \end{Bmatrix} \right| = s(s+0.3) = 0$$

$$s=0 \quad s=-0.3$$

Open loop Poles

$$M = \{B \mid AB\} = \begin{Bmatrix} 0 & 15 \\ 15 & -45 \end{Bmatrix}$$

$$\text{Rank}(M) = 2 \quad \text{"Controllable"}$$

System not in controllable Canonical form.

State-Feedback

$$\begin{aligned} u &= -Kx = -\{K_0 \ K_1\} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} \\ &= -K_0 \theta - K_1 \dot{\theta} \end{aligned}$$

$$\dot{x} = (A - BK)x$$

$$\begin{Bmatrix} 0 & 1 \\ 0 & -0.3 \end{Bmatrix} - \begin{Bmatrix} 0 \\ 15 \end{Bmatrix} \{K_0 \ K_1\}$$

$$\begin{Bmatrix} 0 & 1 \\ 0 & -0.3 \end{Bmatrix} - \begin{Bmatrix} 0 & 0 \\ 15K_0 & 15K_1 \end{Bmatrix} =$$

$$\tilde{A} = \begin{Bmatrix} 0 & 1 \\ -15K_0 & -0.3 - 15K_1 \end{Bmatrix}$$

$$|(sI - \tilde{A})| = 0 \quad \text{Giving closed loop Poles}$$

$$= \left| \begin{Bmatrix} s & 0 \\ 0 & s \end{Bmatrix} - \begin{Bmatrix} 0 & 1 \\ -15K_\theta & -0.3-15K_w \end{Bmatrix} \right| = 0$$

$$\begin{vmatrix} s & -1 \\ 15K_\theta & s+0.3+15K_w \end{vmatrix} = 0$$

$$s(s+0.3+15K_w) + 15K_\theta = 0$$

$$s^2 + 0.3s + 15K_w s + 15K_\theta = 0$$

$$s^2 + (0.3+15K_w)s + 15K_\theta = 0$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

$$2\zeta\omega_n = 0.3 + 15K_w$$

$$\omega_n^2 = 15K_\theta$$

Desired
Characteristics

$$\zeta = 0.7$$

$$\omega_n = 3.24$$

$$K_\theta = 0.7$$

$$K_w = 0.2824$$

Desired Poles
 $-2.25 \pm 2.9917j$
 $-0.25 \pm 2.9327j$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

$$2\zeta\omega_n = 0.3 + 15 K_\omega$$

$$\omega_n^2 = 15 K_\theta$$

Desired Characteristics

$$\zeta = 0.7$$

$$\omega_n = 3.24$$

$$K_\theta = 0.7$$

$$K_\omega = 0.2824$$

Desired Poles
 $(-2.15 \pm 2.391j)$
 $(-2.15 - 2.391j)$

Experimentally, $K_\theta = 0.7$ $K_\omega = 1.5$

Best Response.

State-feedback $\Rightarrow u = -0.7\theta - 0.2824\dot{\theta}$
 Control law Regulator $\theta = 0$

for Tracking $\Rightarrow u = -\{K_\theta K_\omega\} \begin{Bmatrix} \theta - \theta_{ref} \\ \dot{\theta} \end{Bmatrix}$

Set θ_{ref} as desired

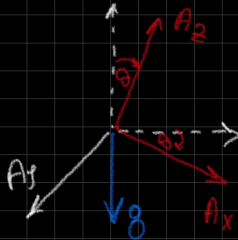
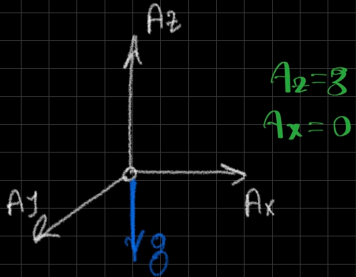
Real-Time Implementation

1) Measurement

MPU Angle $\Rightarrow$
"Obtained from"
Accelerometer

$$A_z = 3 \cos \theta$$

$$A_x = 3 \sin \theta$$



$$\frac{A_x}{A_z} = \tan \theta$$



$$\therefore \theta = \tan^{-1} \left(\frac{A_x}{A_z} \right) [\text{deg}]$$

$$\theta_p = \tan^{-1} \left(\frac{A_x}{A_z} \right) \times \frac{180}{\pi} [\text{rad}]$$

Gyro. access to obtain Speed..

$$\omega = \frac{\Delta \theta_p}{\Delta t} \text{ (rad/s)}$$

$$131 \text{ LSB} = 1^\circ/\text{s}$$

$$\omega_{\text{deg/sec.}} = \frac{G_y}{131}$$

$$G_o = \dots$$

measure velocity
@ Rest to ensure
there's no error

$$\omega = \frac{G_y - G_o}{131}$$

←

$$\omega_{\text{rad/s}} = \left(\frac{G_y - G_{\text{off.}}}{131} \right) \times \frac{\pi}{180}$$

1) Signal Conditioning

- Angle: $\theta_{ref} = \dots \leftarrow \text{Measure @ Startup}$

$$\theta = \theta_m - \theta_{ref} \quad \leftarrow \text{to ensure that } \theta = 0 \text{ (at Startup)}$$

to account for noise of angle measurement.

Measure θ from Gyro $\Rightarrow \theta_{i+1} = \theta_i + \omega \cdot dt$
"Euler's formula"

"Complementary filter" $\theta = \alpha \cdot (\theta + \omega \cdot dt) + (1-\alpha) \theta_{acc}$
 $\alpha = 0.98$

⚡
accounts for drift of gyro reading
& handles high noise of accelerometer.

3) Actuation

Base Throttle

$$u_0 = 1100$$

$$u_1 = u_0 + du$$

$$u_2 = u_0 - du$$

Differential Control

Slow-Rate Limiting

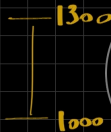
$$|\Delta PWM| \leq \text{Max. allowed Step}$$

$\Delta u_1 \quad \Delta u_2$

Limiting Step to avoid sudden Spikes in PWM

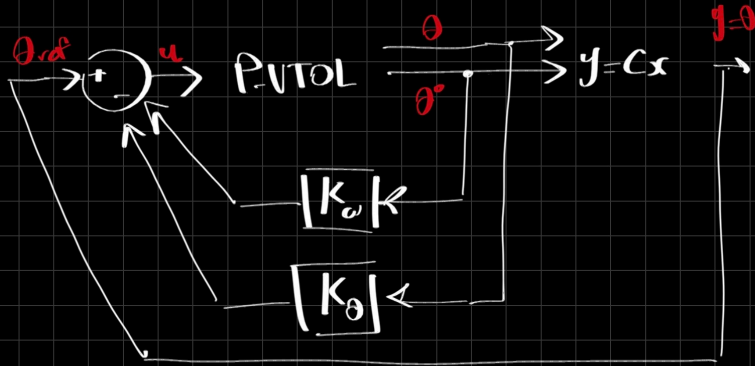
State-feedback

$$du = - \left(\underbrace{K_{\theta} x (\theta - \theta_{ref})}_{\text{Corrects angle}} + \underbrace{K_w \dot{x}}_{\text{Damps Motion}} \right)$$



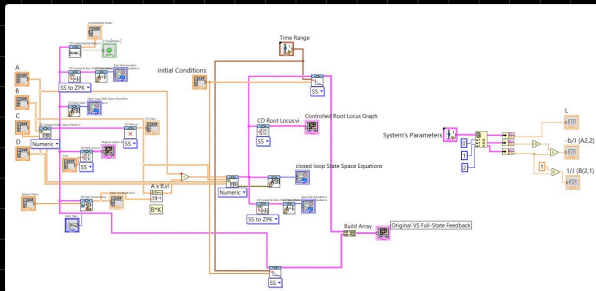
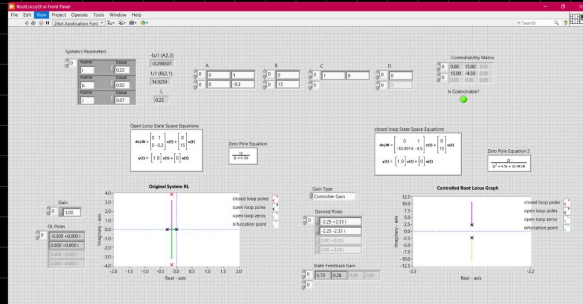
Saturation

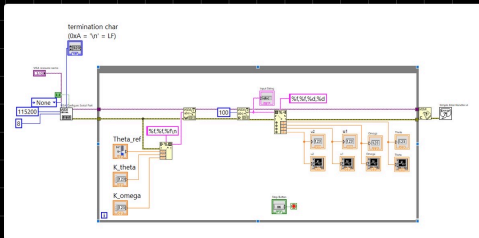
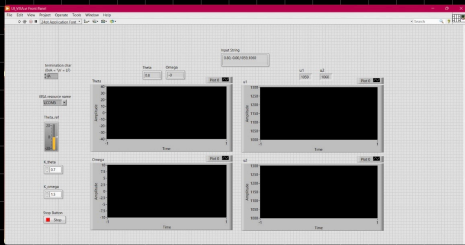
System's Block Diagram



It acts like a PD Controller, but you have two Independent Measurements.

labview VIs





UI for Running experiment

Used NI VISA was Instrumental in
Exchanging Data Between arduino
& LabView (Serial Communication)

- Write θ_{ref} , K_{θ} , K_{ω}
- Read u_1 , u_2 , θ , ω
for Plotting